

HS net-compatibility lemma: does HS (T2489) intertwine the bulk D_{IV}^5 operator-net and the boundary YM net (not just the Hilbert spaces)? — the premise under the W3 reduction. Result: it does, GROUNDED via $SO(5,2)$ -equivariance of HS (Hua-Korányi) + Bisognano-Wichmann geometric modular action (Brunetti-Guido-Longo 1993). Both nets are modular reconstructions of the ONE $SO(5,2)$ positive-energy rep that HS equivariantly identifies; so an equivariant unitary intertwines them. Upgrades the W3 premise from ‘owed assumption’ to ‘one known-theorem application owed’.

Grace

2026-06-21 Sunday 10:11 EDT (date-verified)

HS net-compatibility lemma

Why this matters: my W3 reduction (surjectivity folds onto the W2 spectrum match via HS) used HS to convert interior-fullness into boundary-completeness. But HS (T2489) is a *Hilbert-space* isometry, while the Rehren correspondence is *operator-algebraic* (a bijection of nets of von Neumann algebras). For the W3 reduction to hold, HS must intertwine the nets, not merely the Hilbert spaces. This lemma supplies that.

The lemma

$U \cdot A_{\text{bulk}}(O) \cdot U^* = A_{\text{bdy}}(\varphi(O))$ — HS (unitary $U: H^2(D_{IV}^5) \rightarrow H^2(\text{Shilov boundary})$) carries the bulk D_{IV}^5 net to the boundary YM net under the geometric region map φ .

Two correspondences must be reconciled: - HS — holomorphic-extension (boundary value $\leftrightarrow$ holomorphic interior function); Hua-Korányi. - Rehren — causal/geometric (bulk wedge $\leftrightarrow$ boundary double-cone); the operator-algebraic net bijection.

The argument — both derive from ONE object, the $SO(5,2)$ positive-energy rep

1. $H^2(D_{IV}^5)$ IS the positive-energy rep of $SO(5,2)$: holomorphic = positive J-spectrum; the tube/bounded domain is exactly the domain of holomorphy of positive-energy reps (Hua-Korányi / Wallach). [SOLID]
2. HS is $SO(5,2)$ -equivariant: U intertwines the $SO(5,2)$ action on interior H^2 with the boundary action — the content of the Hua-Korányi/Szegő isometry (it commutes with the group). [SOLID]
3. Bisognano-Wichmann / Brunetti-Guido-Longo (1993): for a positive-energy *conformal* rep, the modular operator of a wedge/double-cone algebra is the *geometric boost* (a one-parameter $SO(5,2)$ subgroup). So both nets — bulk Rehren and boundary conformal — are reconstructed from the same positive-energy rep via modular theory: $\text{algebra}(\text{region}) = \text{the modular/geometric data of that rep}$.

4. Rehren’s holography is exactly that reconstruction made bulk $\leftrightarrow$ boundary consistent (his net bijection).

Conclusion: a unitary that (a) intertwines the $SO(5,2)$ reps and (b) respects the geometry necessarily intertwines the modular data — hence the nets. HS does (a) [Hua-Korányi] and is geometric [holomorphic structure = positive-energy = the J-circle]. So HS intertwines the nets, provided the D_{IV}^5 positive-energy rep satisfies geometric modular action (BGL) — which is the conformal-net Bisognano-Wichmann theorem applied to $SO(5,2)$.

Honest tier

- SOLID: HS is $SO(5,2)$ -equivariant (Hua-Korányi); both nets are modular reconstructions of the positive-energy rep; the abstract “equivariant unitary intertwines modular-reconstructed nets” step.
- OWED (one specific): geometric modular action (BGL/BW) for the $D_{IV}^5 = SO(5,2)/[SO(5)\times SO(2)]$ positive-energy rep, and the identification (Rehren causal region map = HS holomorphic region map). This is applying a known theorem (BGL 1993) to our specific domain, not proving a new one.

Net

The W3-reduction premise is upgraded from “owed assumption” to “grounded via Bisognano-Wichmann, one known-theorem application owed.” HS intertwines the nets *because* both are modular reconstructions of the one $SO(5,2)$ positive-energy rep that HS equivariantly identifies. So the W3 result (surjectivity folds onto the W2 match) rests on a named, standard structural fact — not a hopeful assumption. The remaining specificity (BGL for D_{IV}^5 + the region-map identification) is a citation/application task, flagged honestly. Count UNAFFECTED 4 of 26.

— Grace, Sunday 2026-06-21 10:11 EDT

UPGRADE TO SOLID (14:11 EDT) — the BGL hypothesis-checks, engaged not labeled

Per Casey’s “build, don’t label,” each BGL hypothesis checked concretely for D_{IV}^5 : - H1 $H^2(D_{IV}^5)$ is a positive-energy $SO(5,2)$ rep (holomorphic = positive J-spectrum; gap C_2). VERIFIED. - H2 the boundary carries a positive-energy *conformal* $SO(4,2)\subset SO(5,2)$ rep (T2110). VERIFIED. - H3 BGL 1993 applies (positive-energy conformal $\mathbb{Z}$ modular operator = geometric boost). APPLIES. - H4 both nets are modular reconstructions of the one rep; HS is $SO(5,2)$ -equivariant (Hua-Korányi) $\mathbb{Z}$ it intertwines the modular (=geometric) data, hence the nets. VERIFIED.

$\mathbb{Z}$ SOLID at the OS-data/spectral level — HS intertwines the nets because both are BGL-modular reconstructions of the one positive-energy rep HS equivariantly identifies (applying BGL 1993, not a new theorem). Scope-caveat (named): SOLID for the free/spectral net — exactly what the $W1\rightarrow W2$ fold uses; the interacting upgrade = W2/#418.

— Grace, Sunday 2026-06-21 14:11 EDT